LLM in fabula

GROUP 05



Stefano Riva



Diego Tagliabue



Francesca Trevisan

We all know that what makes a great Lupus in Fabula game session is

a great narrator

But what happens when you don’t have one?

You can probably picture these scenarios:

1.

nobody is able to make a great narrator

you’re engineers, not actors

2.

nobody wants to be the narrator

you want to play and take part in the game, not being a spectator

3.

all game sessions are the same

you always get bored playing consecutive games

We want to change those scenarios

1.

nobody is able to make a great narrator

and that's ok!

you're engineers, not actors

2.

nobody *has* ~~wants~~ to be the narrator

can

you ~~want to~~ play and take part in the game, not being a spectator

3.

all game sessions are ~~the same~~ *unique*

never

you ~~always~~ get bored playing consecutive games

HOW?

a great (LLM) narrator

to spice up your
in-person game sessions

S. Riva, D. Tagliabue, F. Trevisan

project brief

LLM in fabula

your LLM narrator to spice up your game session

S. Riva, D. Tagliabue, F. Trevisan

A.A. 2025/2026

1. FIRST STEP

researching the topic

the game: Lupus in Fabula



1. Social deduction game in which everyone has a secret role
2. The aim is to figure out who’s lying and who’s telling the truth
3. Players debate, accuse and vote to eliminate wolves before they take over the village

state of the art

First use of LLM as narrator AI Dungeon

LLM narrators often evaluated through Dungeons&Dragons

Fable Reborn Virtual Dungeon Master

NAT1 and the summarization module

Lupus in Fabula is used to evaluate models behavior in enviroment full of false information

Werewolf Arena

1. M. Hua and R. Raley, "Playing with unicorns: AI dungeon and citizen NLP," Digital Humanities Quarterly, vol. 14, no. 4, 2020.
2. C. Ang, L. R. Cortel, C. L. Santos, and E. Ong, "Fable reborn: Investigating game-play experience between a human player and a virtual dungeon master," in Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems, CHI EA '23, (New York, NY, USA), Association for Computing Machinery, 2023.
3. B. Patel, P. Shinde, D. H. Joshi, D. S. Rathod, P. R. Vichare, and S. P. Yadav, "Generative ai for dungeons and dragons," in 2025 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI), pp. 1871–1876, 2025.
4. S. Bailis, J. Friedhoff, and F. Chen, "Werewolf arena: A case study in llm evaluation via social deduction," 2024.

state of the art

First use of LLM as narrator AI Dungeon

LLM narrators often evaluated through Dungeons&Dragons

Fable Reborn Virtual Dungeon Master

NAT1 and the summarization module

Lupus in Fabula is used to evaluate models behavior in enviroment full of false information

Werewolf Arena

1. M. Hua and R. Raley, "Playing with unicorns: AI dungeon and citizen NLP," Digital Humanities Quarterly, vol. 14, no. 4, 2020.
2. C. Ang, L. R. Cortel, C. L. Santos, and E. Ong, "Fable reborn: Investigating game-play experience between a human player and a virtual dungeon master," in Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems, CHI EA '23, (New York, NY, USA), Association for Computing Machinery, 2023.
3. B. Patel, P. Shinde, D. H. Joshi, D. S. Rathod, P. R. Vichare, and S. P. Yadav, "Generative ai for dungeons and dragons," in 2025 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI), pp. 1871–1876, 2025.
4. S. Bailis, J. Friedhoff, and F. Chen, "Werewolf arena: A case study in llm evaluation via social deduction," 2024.

state of the art

First use of LLM as narrator AI Dungeon

LLM narrators often evaluated through Dungeons&Dragons

Fable Reborn Virtual Dungeon Master

NAT1 and the summarization module

Lupus in Fabula is used to evaluate models behavior in enviroment full of false information

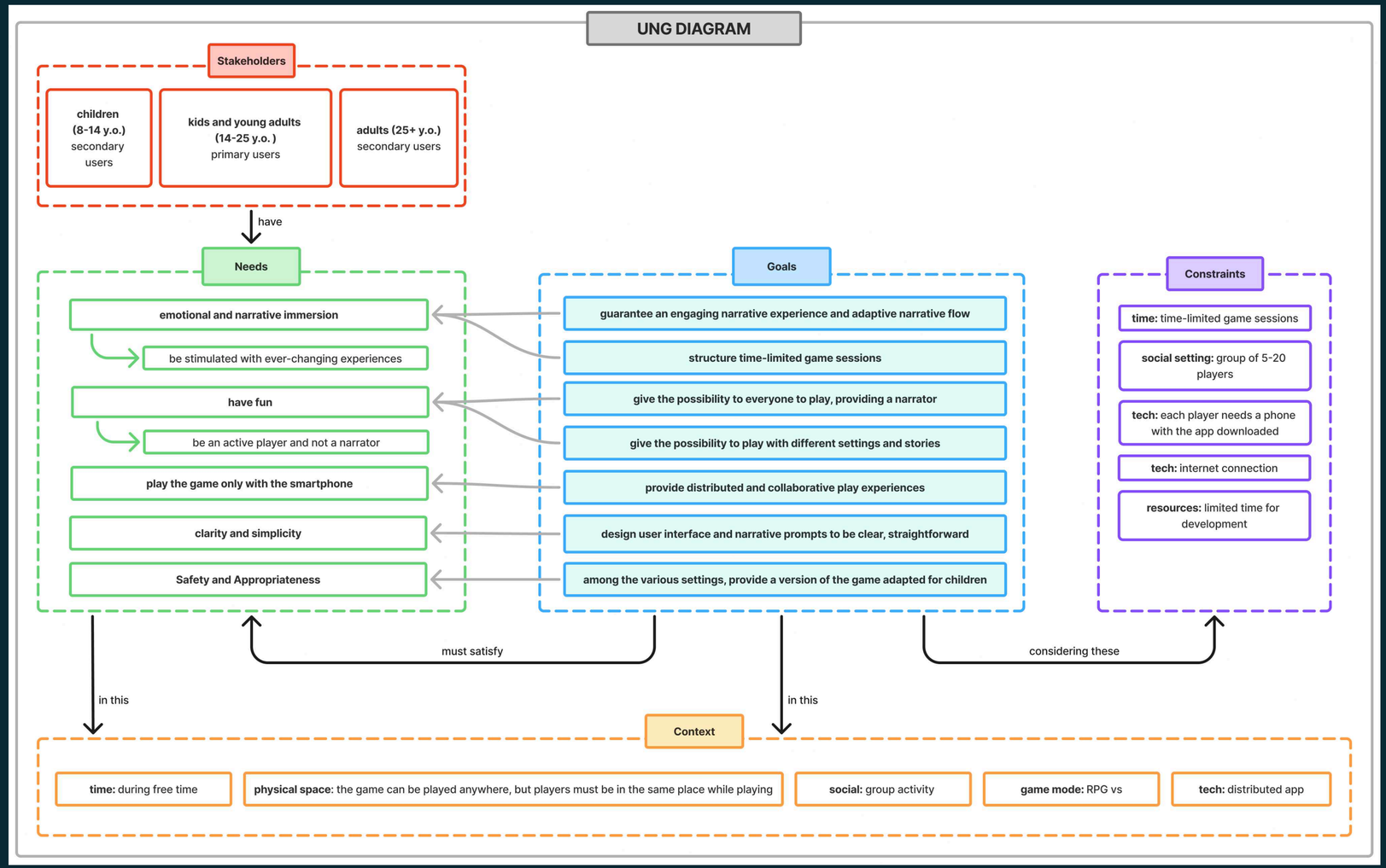
Werewolf Arena

1. M. Hua and R. Raley, "Playing with unicorns: AI dungeon and citizen NLP," Digital Humanities Quarterly, vol. 14, no. 4, 2020.
2. C. Ang, L. R. Cortel, C. L. Santos, and E. Ong, "Fable reborn: Investigating game-play experience between a human player and a virtual dungeon master," in Extended Abstracts of the 2023 CHI Conference on Human Factors in Computing Systems, CHI EA '23, (New York, NY, USA), Association for Computing Machinery, 2023.
3. B. Patel, P. Shinde, D. H. Joshi, D. S. Rathod, P. R. Vichare, and S. P. Yadav, "Generative ai for dungeons and dragons," in 2025 6th International Conference on Data Intelligence and Cognitive Informatics (ICDICI), pp. 1871–1876, 2025.
4. S. Bailis, J. Friedhoff, and F. Chen, "Werewolf arena: A case study in llm evaluation via social deduction," 2024.

2. SECOND STEP

understanding our users

A diagram to guide the project: UNG model



persona definition



children (8-14 y.o)

secondary user

Leo, 10

- Loves being part of the story
- He often plays with his cousins during family gatherings
- He **finds it hard when adults have to explain complex rules**



teenagers/ young adults (15-30 y.o.)

primary user

Matteo, 23

- Busy student which loves party games
- Often **ends up being the narrator instead of playing**.
- He would love if everyone could **participate equally.**



adults (+30 y.o)

secondary user

Marta, 38

- Since having two kids, **time is scarce.**
- She's **looking for games the family can play together that don't require setup or long instructions**

users' needs



children
(8-14 y.o)

secondary user



teen/young adults
(15-30 y.o)

primary user



adults
(+30 y.o)

secondary user

Main needs

- emotional and narrative immersion
- have fun
- play the game only with the smartphone
- clarity and simplicity
- personalized experience

project goals

Main needs

- emotional and narrative immersion
- have fun
- play the game only with the smartphone
- clarity and simplicity
- personalized experience

Main goals

- offer diverse and accessible gameplay
- enhance engagement through adaptive narration
- enable collaborative, time-limited play

3. THIRD STEP

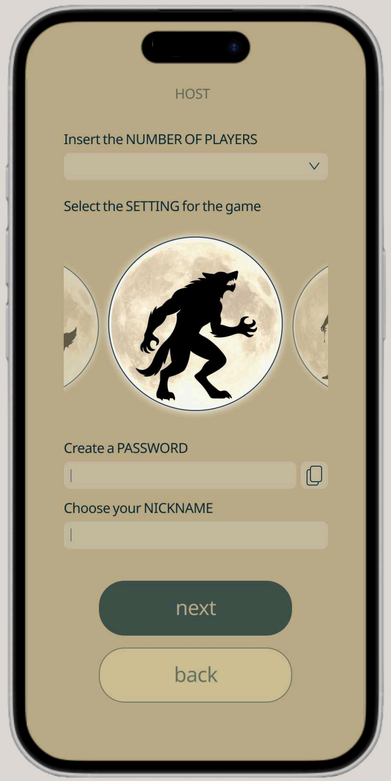
UX design and initial developing

Main characteristic

1. mobile application for in-person sessions
2. LLM narrator and limited time
3. different settings to play



the application



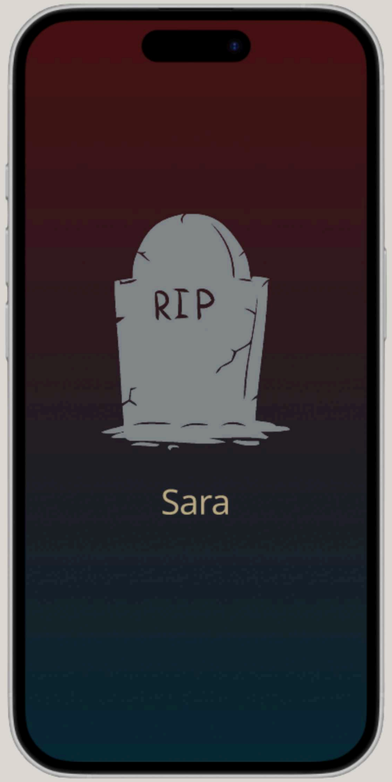
setup page for the host



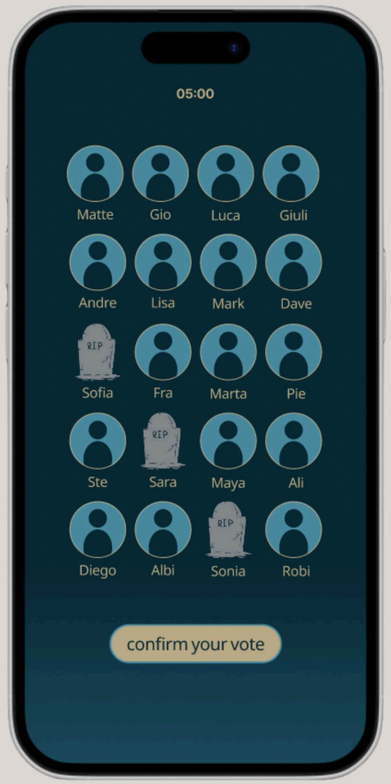
textual output of the narrator (also verbal)



visualization of your role and your “tasks”



death announcement



voting page to select who to address with your action



pool results

4. FOURTH STEP

Technological implementation

architecture

Frontend

Cross-platform mobile app, Flutter

Backend

Django, Phyton

WebSockets, Redis

Narrator

GPT-5.1 model

low latency

real-time synchronization

architecture

Frontend

Cross-platform mobile app, Flutter

Backend

Django, Phyton

WebSockets, Redis

Narrator

GPT-5.1 model

low latency

real-time synchronization

architecture

Frontend

Cross-platform mobile app, Flutter

Backend

Django, Phyton

WebSockets, Redis

Narrator

GPT-5.1 model

low latency

real-time synchronization

architecture

Backend

Django, Phyton

WebSockets, Redis

Frontend

Cross-platform mobile app, Flutter

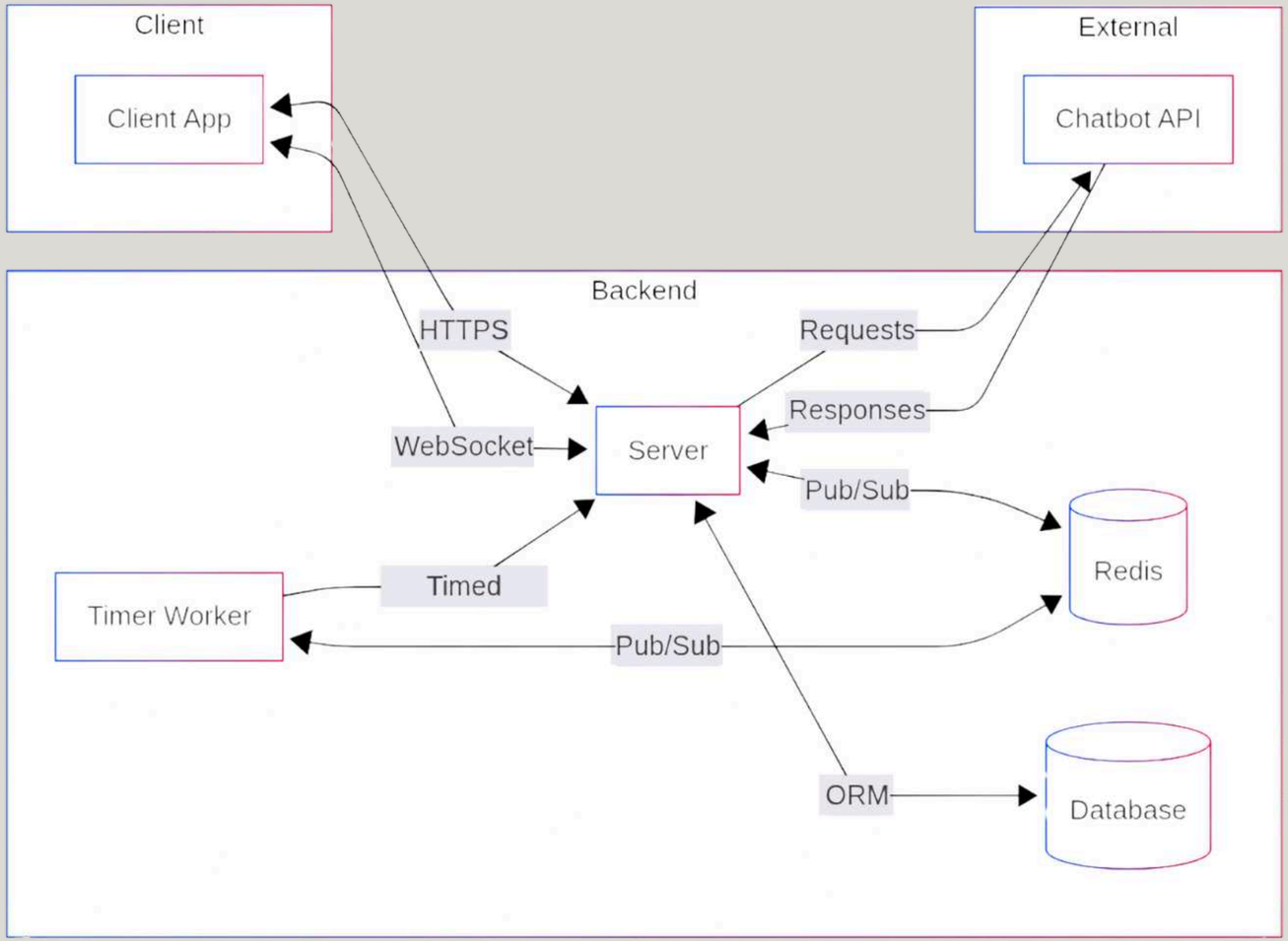
Narrator

GPT-5.1 model

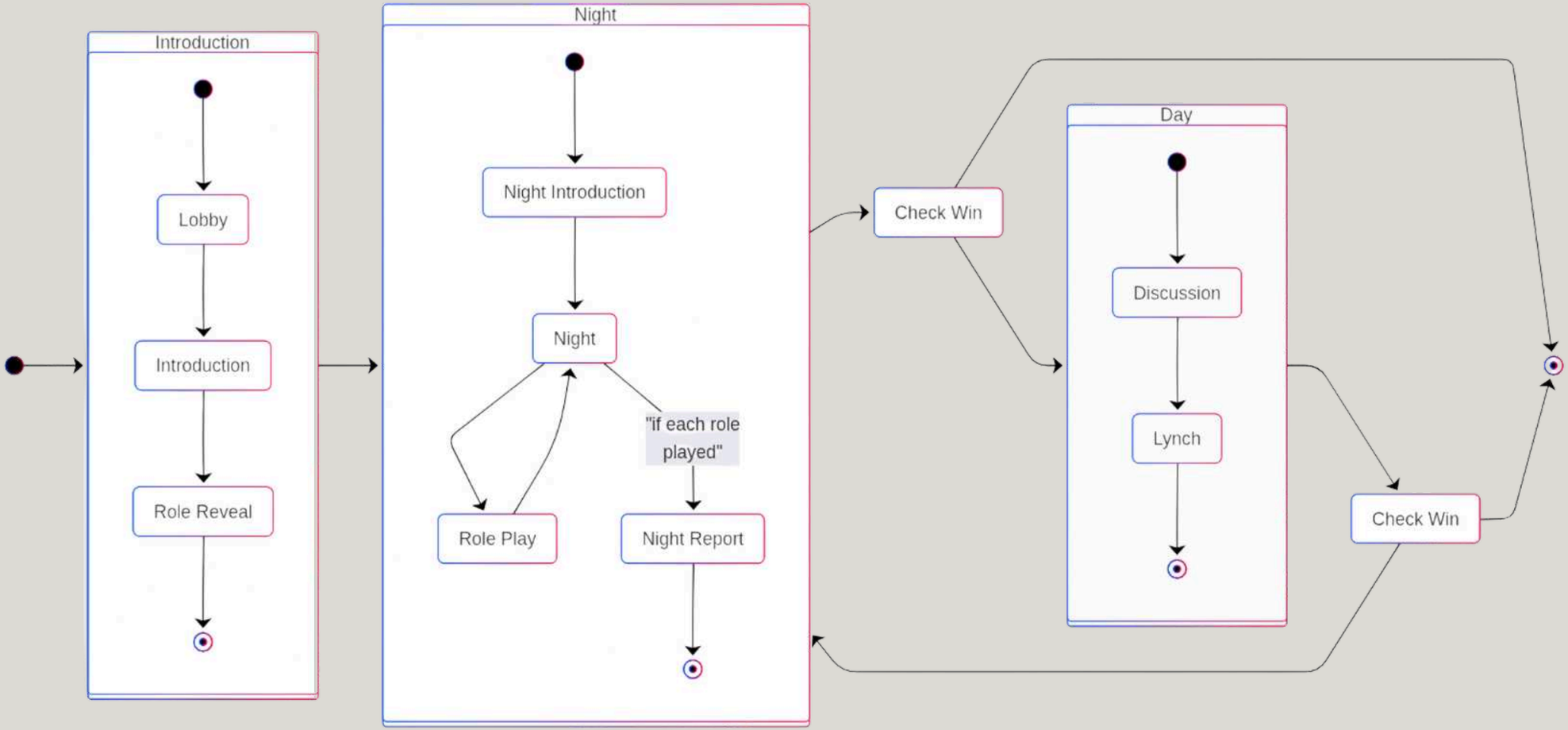
low latency

real-time synchronization

component diagram



state diagram



Now, we all know that what makes a great Lupus in Fabula game session is

LLM in fabula

your LLM narrator to spice up your game session

CONSIDERATIONS

Value proposition and onliness

What makes LLM in Fabula a valid solution?

total participation

adaptive and immersive storytelling

accessibility for all ages

easy setup

***phygital* experience**

enhances the authenticity of human interactions
facilitates in-person connections rather than replace it
technology becomes an ally

What makes LLM in Fabula a valid solution?

total participation

adaptive and immersive storytelli...

accessibility for all ages

easy setup

phygital experience

enhances the authenticity of human interactions
facilitates in-person connections rather than replace it
technology becomes an ally

CONSIDERATIONS

Possible future developments

What's next?

1.

multimodal AI

"Listening Narrator" via Speech-To-Text

2.

new settings

to explore and create new universes

3.

offline resilience

to make the game stable everywhere

THANK YOU!



POLITECNICO
MILANO 1863

LLM in fabula - Final presenation

Group 5: Stefano Riva, Diego Tagliabue, Francesca Trevisan